

Audio Deepfake Detection

Project Name: Audio Deepfake Detection

Sponsor: Timothy Davidson

Date: 10/24/2021

Team Members and Affiliations:

Aidan Gow, The University of Texas at Dallas, ang180000@utdallas.edu

Dan Sutter, The University of Texas at Dallas, dms160130@utdallas.edu

Sagnik Dakshit, The University of Texas at Dallas, sdakshit@utdallas.edu

Vishnu Kunadharaju, The University of Texas at Dallas, vvk160130@utdallas.edu

Mohan Reddy, University of Houston, mputluru@cougarnet.uh.edu

Nihil Patel, University of Houston, npatel59@uh.edu

Keywords: Ensemble learning, Deep learning, Audio spoof detection, Transfer Learning

Short phrase describing the general area within cryptology of the project:

Deepfake technologies, which allow for production of fake images, videos, and audio clips, are attractive to cybercriminals who can use these for disinformation campaigns and fraud. Detecting these deepfakes is a key element of managing this threat.

Keywords: Ensemble learning, Deep learning, Audio spoof detection, Transfer Learning

Project Description: To train and develop a highly accurate machine learning model(s) that can classify if an audio has been deep faked.

Assignment of responsibilities to group members (if group work):

- Aidan Gow and Vishnu Kunadharaju
 - Train a deep learning model on the LFCCs data modality
- Dan Sutter
 - Generate the spectrograms from the 2019 ASVspoof dataset
- Mohan Reddy
 - Generate the LFCCs data modality from the 2019 ASVspoof dataset
- Sagnik Dakshit
 - Train a 2D CNN model using spectrograms that were derived from the 2019 ASVspoof dataset
- Nihil Patel

- Train a 1D CNN model using raw audio features from the 2019 ASVspoof dataset.

Brief list of deliverables:

- Audio deep fake detector(s) machine learning source code
- Mid-progress report
- Mid-progress presentation
- Final project report
- Final PowerPoint presentation

Executive Summary

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The advent of synthetic data from systems such as Generative Adversarial Networks (GAN), Autoencoders, and Genetic algorithms (non-deep learning methods) has inherently bolstered the need for security. Synthetic “fake” audio data, if not detected, can be used to compromise security of not only voice-based systems, but also individual credibility. Furthermore, a part of an authentic audio signal can be forged - having a crucial impact. In this work, we propose a framework to detect deep faked audio signals. Our plan for developing this framework is to explore deep learning-based ensembles using different modalities of data. We also want to explore the use of different audio features instead of traditional audio spectrograms as a means to improve system performance. Our proposed framework components individually serve as end-to-end audio classification, while a non end-to-end ensemble approach that combines these components is expected to improve performance over time.

Motivation

Problem Statement: In this project, we seek to answer the following question. Can we develop a robust machine learning model that can accurately classify audio clips that contain significant background noise as deepfake or original? We are tackling this question by proposing a framework to identify synthetic audio signals generated logically that can potentially compromise security. Currently, research in this area is in its early stages. As a result, the most difficult challenge in our project is experimenting and training machine learning models with various combinations of data modalities in order to determine which modalities and ML models will be the most effective at classifying synthetic audio.

Purpose Statement: The goal of our project is to create a machine learning model or models that can accurately classify whether a piece of audio is synthetic (deep faked) or genuine. Our solution will help existing fake media detectors identify and combat mass-produced highly realistic synthetic content. Furthermore, it has the secondary goal of protecting our society and democracy by safeguarding the information channels on which we rely upon when forming our beliefs and opinions.

In current times, technology is growing at an exponential, almost uncontrollable, rate. We can now mask and change parts and functions of the human body - one of these being the voice. People's voices can be changed and warped for various purposes, some of which could be fueled by malicious intent. This is scary because these deepfakes could steal and impersonate someone else's voice to an almost unnervingly realistic extent. The idea behind this project would be to logically detect these changes and classify whether a piece of audio is a deepfake or not. Any normal person would lack the sufficient knowledge to be able to detect and decipher these changes. With our project, we hope to achieve a level of protection against misinformation that would prevent this technology from being able to attack the average citizen. As deepfake technology becomes more readily available to the public, it will be easier to spread this fake content at a much faster rate. With the proliferation of such media, it will be increasingly difficult for citizens to be accurately informed on critical issues in our country. Our solution seeks to prevent this scenario from becoming a reality. The most difficult challenge we face is the fact that research in this area is still in its early stages. As a result, it is our responsibility to experiment and train multiple machine learning models on various data modalities to determine which modality and ML model(s) will produce the best classifier. This set of experiments represents the work that must be completed in order to develop a powerful machine learning model(s) capable of accurately detecting synthetic audio. Once we have created such a model, we will be able to use it in conjunction with existing fake media detectors to combat the rise in

mass-produced highly realistic synthetic content. If we achieve our goal, our solution will help to protect our society and democracy by protecting the information channels that Americans rely on when forming their beliefs and opinions. On a more personal note, we, as a team, hope to further our knowledge of deep learning systems, and grow as developers. As young and impressionable developers, we want to be able to add relevant and meaningful projects to our repertoire. This project will allow us to grow our backgrounds in both cyber security and machine learning.

Previous Work/ Literature Review

In this section, we discuss some of the relevant work in audio deepfake and spoof detection non-exhaustively. The work on deep fake audio detection is fairly nascent. The problem has been posed in different forms like audio spoof detection and automatic speaker verification. The most common approach is to convert audio signals to the frequency domain as spectrograms for 2D deep learning classifiers [1] among others such as using raw waves and applying Short-time Fourier Transform (STFT). The 2D spectrograms are adequately representative of additive attacks, pitch/frequency modulations and replacement attacks [3]. While 2D convolutional neural network (CNN) based approaches have achieved over 95% accuracy in spoof detection using transfer learning, the authors verify that a combination of modalities of raw waveform and 2D spectrograms perform superior than either individually [2]. Chettri, Bhusan et. al [4] discuss posing the problem in terms of automatic speaker verification (ASV). The authors exhaustively evaluate ensembles of deep and traditional machine learning models such as logistic regression, Support Vector Machines, and Gaussian Mixture models to demonstrate the superior performance of ensemble methods. Chen et. al [6] proposed a LDA-based lower dimensional embedding paradigm for author verification for generalized author verification across datasets. Spec-ResNet [5], compared to other discussed approaches, proposes a superior steganography scheme in spectrograms to secure audio signals. Their classification model achieves accuracies in the range of 82-93% on various datasets. Joan et. al [7] proposed an end-to-end method using self-attention networks with temporal pooling to achieve high accuracy. Clearly, the previous efforts have made great strides in tackling this issue, but there is room for improvement. Upon review of the previous literature, it became apparent that other audio features such as amplitude, pitch, and structural patterns were overlooked in favor of spectrograms. For our solution, we seek to leverage both these overlooked features and spectrograms in an ensemble model to classify synthetic and original audio inputs. Additionally, we want to focus on microphone forensics and environment/background noise subtraction to improve classification accuracy. We seek to develop a more robust machine learning solution that can accurately classify audios even with the presence of various background noises.

Specific Aims

Plan Overview: It is established that spectrograms are frequency features of any waveform. The use of only spectrograms in deep learning restricts learning to the frequency space. The benefits of other audio features such as amplitude, pitch, and structural pattern spaces are overlooked. We want to explore classification of synthetic and original audio inputs by leveraging knowledge of both spectrograms and 1D audio signals. 1D deep learning models are relatively lightweight and allow near real-time system performance, which is critical to security tasks. Deep learning models learn the frequency and structural features of an input signal better from 1D signals [8] than from spectrograms where any structural features are lost. We will leverage existing knowledge by transfer learning to develop these individual models over training from scratch given our limited resources. We expect the ensemble of modalities to perform better while allowing us in turn to explore the fallacies in other audio-based security systems in future. We also will explore other possible extracted features that might yield superior performance over the traditional spectrograms and raw signals. The deliverables include the python source code for the machine learning model(s), the final presentation report, and the final presentation to our sponsor and the InSure organization. The mid-progress presentation and report deliverables have been completed up to this point.

Plan

Plan Overview: If feasible, our team will be attending an Automatic Speaker Verification Spoofing and Countermeasures (ASVspoof) workshop, on 09/16/2021, that will go over current methods to create audio deepfakes. We will gain a better understanding of how audio deep fakes are currently created, as well as have access to the data needed to extract the various audio features and train our machine learning models, through this workshop. We will start by identifying, understanding, and processing our desired datasets. A large portion of our efforts will be allocated to the selection of deep learning architectures, plus training and optimizing the models individually and as an ensemble. We want to explore new features for efficient model performance. When we arrive at the implementation stage of the plan, we will be leveraging the existing deep learning libraries of PyTorch and TensorFlow. In addition, the data we will use originates from the ASVspoof challenge. This challenge contains data of deepfake audio pieces that contain background noise. After creation of the detector, we will evaluate the model using different metrics and if required perform the necessary tweaks to improve our detector's performance. Additionally, we will test our detector against the synthetic audio from the *Roadrunner* (2021) documentary. The deliverables include the python source code for the machine learning model(s), the final presentation report, and the final presentation to our sponsor and the InSure organization. The mid-progress presentation and report deliverables have been completed up to this point.

Detailed Plan:

1. Work already done

a. Generation of spectrograms from 2019 ASVspoof challenge.

i. Brief Task Description:

1. We took the existing audio data from the 2019 ASVspoof challenge and derived the corresponding spectrograms.

ii. Task Headline:

1. The results were the generated spectrograms which were derived from the 2019 ASVspoof data.

iii. Detailed Task Description:

1. A Python script was written to take the FLAC audio signals and generate the corresponding spectrograms.

iv. Task Outcomes

1. The spectrograms were generated and now can be used to train a 2D CNN model.

b. Training a 2D CNN model on spectrograms derived from 2019 ASVspoof challenge.

i. Brief Task Description:

1. On the derived spectrograms from the 2019 ASVspoof challenge dataset, a 2D CNN deep learning model was trained.

ii. Task Headline:

1. The task of training the 2D CNN was successful itself, however, the model gave us suspicious metrics upon completion of the training phase.

iii. Detailed Task Description:

1. The team leveraged pytorch/tensorflow to develop the 2D CNN and train it on the spectrograms derived from the 2019 ASVspoof challenge.

iv. Task Outcomes:

1. As mentioned previously, the model gave us suspicious metrics upon completion of the training phase. Upon further investigation it was discovered there was a major class imbalance in the 2019 dataset. To be more precise, there are more AI labels (deep faked audio data) compared to bonafide labels (genuine audio). Specifically, there are over 22296 spoofed labels vs 2548 bonafide labels. This caused our trained 2D CNN to label any new audio input as spoofed or deep faked.

2. Work currently in progress

a. Generation of LFCCs from 2019/2021 ASVspoof challenge

i. Brief Task Description

1. Deriving the LFCC data modality from the audio signals in the 2019 ASVspoof dataset.

ii. Task Headline

1. The task was recently successful and completed. The result is the LFCC for each audio signal and its corresponding bonafide or AI label.

iii. Detailed Task Description

1. A python script was written which derived the LFCCs for each audio signal and stored them in a numpy array.

iv. Task Outcomes

1. We now have the LFCC data modality, and it can be used to train various deep learning models and determine how effective it is in identifying synthetic audio.

b. Training a 1D CNN model on raw signal features

i. Brief Task Description

1. We are training a separate 1D CNN model based on the raw signal features.

ii. Task Headline

1. We expect to have a separate machine learning model that is trained specifically on the raw signal features. Our goal is to incorporate this 1D CNN model with other machine learning models trained on different data modalities to create an ensemble of machine learning classifiers. We are expecting this ensemble of machine learning classifiers to yield us a high accuracy rate.

iii. Detailed Task Description

1. This is still being developed. As of right now the pytorch library is being used.

iv. Task Outcomes

1. N/A yet.

3. Work to be carried out

a. Training a machine learning model on the LFCCs data modality

i. Brief Task Description

1. Select and train another deep learning model on the LFCCs that were generated from the 2019 ASVspoof dataset.

ii. Task Headline

1. We expect to have a separate machine learning model that is trained specifically on the LFCCs. Our goal is to incorporate this ML model with other machine learning models trained on different data modalities to create an ensemble of machine learning classifiers to yield us a high accuracy rate.
 - iii. Detailed Task Description
 1. We have yet to arrive at this stage.
 - iv. Task Outcomes
 1. We have yet to arrive at this stage.
 - b. Resolving the class imbalance in 2019 ASVspoof challenge
 - i. Brief Task Description
 1. Resolving the class imbalance between the bonafide and AI labeled data points in the 2019 ASVspoof challenge.
 - ii. Task Headline
 1. We expect to see a more balanced dataset between the two class labels. Currently there are 22296 AI labeled data (spoofed audio) vs 2548 bonafide labeled data (genuine audio).
 - iii. Detailed Task Description
 1. We are currently looking into an under sampling approach to resolve this class imbalance. If we decide to take this route, we will keep 2548 bonafide labeled data and randomly sample 2548 AI labeled data points into a new dataset. We have also raised this issue to our project sponsors. In the next weekly meeting, we will ask if they have any suggestions on how to resolve this imbalance.
 - iv. Task Outcomes
 1. We have yet to arrive at this stage.
 - c. Compare results between different modalities on deep learning architectures and pre-processing background noise out of the data.
 - i. Brief Task Description
 1. We are experimenting and training different machine learning models based on various data modalities to see which ones would yield us the best classification results.
 - ii. Task Headline
 1. At the end of this phase, we will select the best individual machine learning models that were trained on different data modalities and group them into an ensemble. This ensemble

of various machine learning models should yield a higher accuracy score.

- iii. Detailed Task Description
 - 1. We are not at this stage currently.
- iv. Task Outcomes
 - 1. We are not at this stage currently.

Deliverables

The primarily deliverables can be listed as:

- The audio deepfake detector's source code
- Mid-progress presentation (PowerPoint)
- Mid-progress report
- Final project report
 - We will specifically highlight how accurately our detector classified synthetic and original audio pieces.
- Final presentation (PowerPoint)
 - The presentation will be a summary explaining the following:
 - Our approach
 - Preprocessing of data
 - Feature selection
 - Model training
 - Issues that were encountered and how they were solved
 - Final results

Results To Date

- The most crucial result to date is the class imbalance located in the 2019 ASVspoof dataset. If this issue is not resolved this could force our team to pursue a new direction in our project. To recap, the 2019 ASVspoof dataset has 22296 AI labeled data vs 2548 bonafide labeled data. AI labeled data indicate all the audio signals that are deep faked whereas the bonafide data indicates all the audio signals that are genuine. Due to training the 2D CNN on this significant imbalance when it receives a new audio as input it will just classify it has deep faked. This is an issue that is inherent with the dataset itself. As mentioned previously, we are exploring certain approaches that can resolve this issue. One such approach is under sampling where we keep the 2548 bonafide data and randomly sample 2548 AI labeled data. This strategy should yield us a balanced dataset, but we are throwing away a lot of data from the current dataset. Until this situation is resolved we cannot really make significant progress because all future machine learning models will suffer with the same issue if this imbalance

persists. This issue was flagged and alerted to our project sponsors. In our next weekly meeting, we are planning on asking for suggestions on how to resolve this issue. In the worst case we might need to generate more bonafide data or abandon the 2019 ASVspoof data and turn to the 2021 ASVspoof dataset.

Issues / Challenges

- One of the first problems we ran into was a lack of extensive computational power to derive the various data modalities (spectrograms, lfccs, etc) and train the deep learning models. This was one of the issues we forecasted in our proposal at the start of the semester. To derive the spectrograms and train the 2D CNN, we were initially limited to our personal computing devices. As a result, generating the required items took a significant amount of time. Fortunately, a member of our team was able to obtain a Ryzen 5 3600 and a 2060 Super device. Access to these devices greatly increased our computational resources and significantly reduced the time to derive and train different data modalities and models.
- The significant class imbalance in the 2019 ASVspoof challenge was an unanticipated occurrence. As previously stated, our team was deriving spectrograms from audio signals in order to train the 2D CNN deep learning model. We examined the model's metrics after training it and discovered some unusual results. We discovered a major class imbalance between genuine and AI labels after further investigation. Genuine labels correspond to authentic audio signals, while AI labels correspond to deep fakes signals. There are only 2548 genuine labeled data versus over 22296 AI labeled data. Due to training the 2D CNN on this imbalance when the model is given a new audio input it will just classify it as a synthetic (deep faked) audio. This class imbalance needs to be resolved before any meaningful progress can be made. Currently, we are looking into performing an under sampling of the data where we keep all the 2548 bonafide labeled data and randomly sample 2548 AI labeled data. This issue has been brought up to our sponsors and we will be discussing other approaches to resolve the class imbalance. If none of these work then we would need to utilize the 2021 dataset from the ASVspoof challenge to train the deep learning models.
- Another challenge our team has come across is the lack of transfer learning for 1D CNN models. In our plan, we wanted to train a 1D CNN model on the raw features of an audio signal. The issue is, unlike 2D CNN models such as ResNets and InceptionNets, there are no pre-trained models for 1D CNN. This is another issue we forecasted at the beginning of the semester. The lack of transferring learning will just require more time on our part to develop, train, and fine-tune the 1D CNN. This is something we have to be wary of as our semester is drawing to a close.

- The presence of background noise in the audio data that we are training the deep learning models on. As a result, we will likely need to train our models to detect synthetic audio in spite of the noise or preprocess the data and strip the background noise in the beginning.

Bibliography

- [1] R. L. M. A. P. C. Wijethunga, D. M. K. Matheesha, A. A. Noman, K. H. V. T. A. De Silva, M. Tissera and L. Rupasinghe, "Deepfake Audio Detection: A Deep Learning Based Solution for Group Conversations," *2020 2nd International Conference on Advancements in Computing (ICAC)*, 2020, pp. 192-197, doi: 10.1109/ICAC51239.2020.9357161.
- [2] Krishnan, Anandha, and Don Caeiro. "Detection Of Alterations In Audio Files Using Spectrograph Analysis."
- [3]<https://securityintelligence.com/articles/why-deepfake-audio-technology-is-a-realthreat-to-enterprise-security/> by Mark Stone on 23rd August, 2019.
- [4] Chettri, Bhusan, et al. "Ensemble models for spoofing detection in automatic speaker verification." *arXiv preprint arXiv:1904.04589* (2019).
- [5] Ren, Yanzhen, et al. "Spec-resnet: a general audio steganalysis scheme based on deep residual network of spectrogram." *arXiv preprint arXiv:1901.06838* (2019).
- [6] Chen, T., & Khoury, E. (2021, January). Spoofprint: A New Paradigm for Spoofing Attacks Detection. In *2021 IEEE Spoken Language Technology Workshop (SLT)* (pp. 538-543). IEEE.
- [7] Monteiro, J., Alam, J., & Falk, T. H. (2020). Generalized end-to-end detection of spoofing attacks to automatic speaker recognizers. *Computer Speech & Language*, 63, 101096.
- [8] Maweu, B. M., Dakshit, S., Shamsuddin, R., & Prabhakaran, B. (2021). CEFes: A CNN Explainable Framework for ECG Signals. *Artificial Intelligence in Medicine*, 115, 102059.

Biographical Sketch of Investigators

- **Sagnik Dakshit:** Sagnik is a 4th year PhD student working on Explainable Deep Learning for healthcare. He has worked extensively on Machine Learning and Deep Learning projects with currently 3 journal publications and 2 papers in review. He has also interned with Bell Labs, HP Inc. and IBM India on various ML, DL projects. You can find more about me at www.sagnikdakshit.com
- **Aidan Gow:** I am an Undergraduate student in Computer Science at UT Dallas in their fast track program. This is my second semester taking graduate courses with a track in Intelligent Systems. I work as a Linux System Administrator for DOD accounts as a contractor with certifications in both CompTia Linux+ and Security+. Although I would like to branch more into Software Development and machine learning. I have been studying in the computer science field since my sophomore year of high school. As far as coursework goes, I have (or currently) taken machine learning, web programming languages, and all the required upper level undergraduate requirements at UTD. These include automata theory, operating systems, organization of programming languages, and much more.

Although I may be lacking in extensive experience, I think this project would be a great opportunity for me to grow. My machine learning program has gone into extensive detail over the masking and manipulation of images and other media using ML algorithms.

- **Dan Sutter:** I am a Master's student in Computer Science at UT Dallas, graduating this semester. My track is in Information Assurance. I have taken a variety of computer security and artificial intelligence courses at both the undergraduate and graduate levels, including Intro to Cryptography, Network Security, Artificial Intelligence, Machine Learning, and Computer Vision. I am CompTIA Security+ certified. I am also in the CyberCorps: Scholarship for Service program (SFS), which requires participation in UTD Computer Security Group student club meetings and Capture-the-Flag competitions. Additionally, I read and gave a short presentation on a paper about detecting deepfake-generated images (*Attributing and Detecting Fake Images Generated by Known GANs*, Joslin and Hao) to other SFS students as part of the program.
- **Mohan Reddy:** I'm a master's student at UH in Computer Science. I have worked as a Data Analyst for 3 years and also as a research assistant at NSS Lab, UH with 2 papers under review. Most of the subjects. I am more interested in interdisciplinary fields and applications like ML, AI, NLP, Cybersecurity. I have worked on a few Cyber Security Projects such as Network Intrusion in CAN, Cyberbully detection using NLP methods, Analysis of Network traffic pattern using NetworkX and matplotlib library in Python, etc.
- **Nihil Patel:** I'm a Masters' Student majoring in Computer Science with a specialty in data analytics. I've taken classes in big data, data mining, machine learning, artificial intelligence, computer vision, calculus, and linear algebra at the undergraduate and graduate level. My research background is almost exclusively in deep learning, and I have participated in several internships in the field. Currently, I am conducting a remote internship with the Neuronal Intelligence laboratory in Tübingen Germany, where we are partnering with the startup firm StriaTech to develop a deep learning algorithm that uses the DeepLabCut 3D pose estimation software that will track the movement of a person's skeleton in a video setting and check for behavior consistent with an epileptic seizure.
- **Vishnu Kunadharaju:** I am a graduate student, studying computer science with a focus in information assurance, at the University of Texas at Dallas (UT Dallas). During my time as a graduate student I have taken the following relevant course: machine learning. Upon reviewing the project description, it is clearly evident that machine learning will play a critical role to determine if a piece of media has been engineered by deepfake technologies. As such, my prior background in this field, derived from the machine learning course, will be quite useful as I progress further into the project. In addition, I am well adept with the

python programming language. I have used it extensively for my ML projects and the various automation tools that I developed at my past two internships.

Schedule

Milestone	From	To	Comment
Literature Review of existing methodologies	09/10/21	10/29/21	We will continue to review literature, as needed, the further we progress into the project.
Study of Audio deepfaking methods and the data transformation of audio signals into various components	09/17/21	10/01/21	
Gaining access to Ryzen 5 3600 and a 2060 Super	10/10/2021	10/16/2021	
Generating the LFCCs	10/17/2021	10/19/2021	
Determining Solution Framework and Tools required by analysing suitable deep learning architectures	10/01/21	10/15/21	
Prototype Implementation of Audio Deepfake Detector and exploring ensemble approaches	10/15/21	10/29/21	Mid-Progress: Oct 21st and 22nd
Resolve class imbalance	10/24/21	10/29/21	
Train ML model on the LFCC data modality	10/24/21	10/31/21	
Train 1D CNN model on raw audio features	10/24/21	11/4/2021	
Results Documentation and Reporting	11/12/21	11/26/21	
Final Report & Powerpoint Presentation	11/26/21	12/10/21	Due Dec 9th and 10th

Broader Impact

Cybercriminals can use deepfake technologies in order to create disinformation campaigns and fraud. These technologies will allow for production of fake images, videos, and audio clips. In a broader context of the project, we want to increase both the cyber awareness and security for the average citizen. Deepfakes, as a whole, could be a threat to democracy as our political leaders' own voices could be used to make fake speeches. In current times, with the Covid-19 pandemic, technology is used to communicate more than ever, even more so for political figures and other influencers. These influencers can be mimicked and their resulting reputations will be abused by

cybercriminals. This tool will ensure that, in a logical aspect, we can detect when an AI manipulates audiovisual content. It will be as easy as running a script on a piece of audio in order to detect deepfakes, making it easy and accessible for both the public and private sector. We also will be able to inform people who want to learn more about audio deepfakes using our algorithm about what kind of changes or discrepancies are able to give away a deepfake. This project will enhance the state-of-the-art in cybersecurity by increasing awareness and protection against scammers who utilize deepfaked audio. We can remove it as an option for attackers, although in the future, this could lead to escalation. To elaborate, with the introduction of deepfake detection, it could force attackers to create better deepfakes. But this is merely speculation. In a perfect world, this tool would completely remove deepfakes as an option. We hope to protect our everyday citizens, our influencers, and our political figures for our national and individual security.